1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In HER2-positive breast cancers that express ER at 1–10% and have Ki-67 ≥10%, a high stromal-to-tumor ratio (TSR) identifies a desmoplastic, immune-excluded microenvironment in which trastuzumab-based therapy yields diminished tumor response due to reduced antibody penetration and impaired effector cell engagement, resulting in a lower objective response rate (ORR) compared to ER 1–10% and Ki-67 ≥10% tumors with low TSR. Mechanistic basis includes: (a) dense ECM and activated stroma creating physical and biochemical barriers to trastuzumab diffusion and Fc-mediated ADCC; (b) potential shift in tumor dependence away from HER2 signaling in an ER-low, highly proliferative state; (c) altered immune contexture (reduced CD8+ T cell infiltration, NK cell activity) associated with high TSR. This triad (ER 1–10% + Ki-67 ≥10% + high TSR) defines a distinct, less HER2-addicted phenotype with reduced responsiveness to trastuzumab-containing regimens. The effect should be detectable after adjusting for HER2 amplification level and established prognostic signatures (e.g., LobSig, RS). Secondary modifiers (e.g., LINE-1 methylation, HbA1c) may modulate magnitude but are not essential to the core mechanism. The predictive model will integrate ER status, Ki-67, TSR, and stromal/immune signatures to stratify patients for HER2-targeted therapy intensity and to identify candidates for combination approaches targeting the stroma or boosting immune effector function.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: retrospective clinical cohorts with quantified TSR, ER, Ki-67, and response data;

Prediction 2: mechanistic correlative studies assessing intratumoral trastuzumab levels and immune-effector markers (CD8, NK markers, FcγR engagement) and stromal signatures;

Prediction 3: preclinical models (PDX or organoids with controlled stromal content) to test trastuzumab efficacy and the impact of stromal modulation. If validated, TSR-guided stratification could inform treatment intensification or combination strategies (e.g., stromal remodeling agents, immune activation) for the high-TSR ER1–10% Ki-67≥10% subset to improve outcomes.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions